

privacy constraints. In addition, differential privacy mechanisms introduce noise that accumulates across dimensions, further degrading the quality of the learned representation. As a result, high-dimensional features are more prone to distortion, leading to reduced predictive performance in downstream tasks.

Impact of Degree of Imbalance. To assess the impact of data distribution imbalance among parties on model utility, we vary the data proportion across three parties, where settings like $[0.8, 0.1, 0.1]$ indicate that 80% of the data is held by one party and only 10% by each of the others. The results are shown in Table V-E. As the table reveals, model performance improves as the data distribution becomes more balanced, with the best results achieved under uniform allocation. This trend is consistent across all methods, particularly in *MASTER*, which shows a clear performance gain from 0.654 to 0.846 as the distribution transitions from highly imbalanced to even. This effect can be explained by the fact that severe imbalance limits the diversity and representativeness of local data, especially for minority parties. In privacy-preserving settings, such asymmetry can amplify overfitting or undertraining on low-resource participants. A balanced data distribution ensures that each local model contributes meaningfully to the global objective, leading to better alignment in representation learning and improved generalization of the generated data.

VI. RELATED WORK

A. Differentially Private Single-Party Data Publishing

Tang et al.[16] addressed vertically partitioned data through a DP mechanism balancing privacy with usability. Yao et al.[17] focused on trajectory data, integrating sensitive label protection within GAN-based generation. Jiang et al.[18] investigated cost-effectiveness of synthetic data production under Rényi differential privacy using a pricing model for GAN-based generators. Nieminen et al.[19] examined privacy amplification via subsampling in tabular data generation using GANs, emphasizing empirical utility-privacy trade-offs. Guan et al.[10] introduced diffusion-based privacy-preserving educational data generation, while Park et al.[11] applied diffusion models to differentially private image classification. Jiang et al.[25] analyzed differentially private synthetic data utility through measurement error analysis. Fuentes et al.[7] proposed Joint Selection, adaptively incorporating public data into private synthetic data generation. Cai et al.[26] developed a differentially private synthetic data generation framework for robust information fusion across multiple sources, enhancing data utility while maintaining formal privacy guarantees. Zhang et al.[27] introduced SPINML for generating customized synthetic data tailored to specific machine learning tasks under DP. Padariya [8] presented a privacy-preserving framework for synthetic dataset generation using generative models with general applicability across different data modalities.

B. Differentially Private Multi-Party Data Publishing

For tabular data, Cheng et al. [28] explored high-dimensional data publishing under differential privacy in multi-party settings, offering a strategy for protecting sensitive information in collaborative environments. Xia et al. [29] investigated heterogeneous differential privacy mechanisms for vertically partitioned databases, improving privacy in distributed data sharing. Wang et al. [20] proposed a GAN-based method for secure multi-party data exchange, ensuring differential privacy while preserving data utility for joint analysis. Li et al. [30] introduced a local differential privacy mechanism for trajectory statistics release, applying a water-filling optimization approach to balance privacy

and accuracy. Zhang et al. [12] employed the Bregman divergence to enhance inter-regional connectivity.

For sequential data, Tang et al. [31] proposed a privacy-preserving framework for sequential information shared among multiple parties. They further extended this line of work in [32] by introducing a personalized differential privacy method for marginal release in multi-party contexts. To improve data utility, Tang et al. [1] enhanced their previous work by providing a detailed framework for publishing time-series data under differential privacy, addressing the trade-off between privacy and utility across contributors. Liu et al. [33] developed a method for releasing multi-dimensional sequential data while satisfying local differential privacy, aiming to protect individuals' data at the collection stage while maintaining analytical effectiveness.

In summary, existing approaches face several limitations in the context of multi-party data publishing. Many focus exclusively on single-party scenarios, lacking practical mechanisms for decentralized settings. Others target specific data modalities—such as image data in diffusion models—which are often inefficient, less scalable, or poorly aligned with the diverse and structured nature of data in real-world multi-party applications where both privacy and utility are essential. Moreover, current methods typically model data distributions from a single perspective, leading to significant divergence between the distributions of real and generated samples under DP. This one-sided fitting strategy fails to capture the heterogeneous characteristics across parties, resulting in degraded utility when applied in collaborative environments.

VII. CONCLUSION AND FUTURE WORK

In this paper, we present *MASTER*, which is a weight-aware bidirectional GAN framework that effectively addresses the limitations of existing differentially private GAN in terms of sample quality, diversity, and training stability for horizontal multi-party data publishing. By integrating private discriminators, a dual-generator architecture, and an alternating optimization strategy, *MASTER* enables robust training and efficient fusion with strong theoretical and empirical support.

Future work will focus on extending *MASTER* to vertical federated learning scenarios, where parties hold different features of the same sample set. This introduces several new challenges: (1) Joint modeling across heterogeneous feature spaces requires effective semantic alignment between generators and discriminators; (2) Imbalanced feature dimensions may lead to uneven generation quality, calling for dynamic coordination mechanisms; (3) Privacy protection must be adapted to feature-level perturbations and cross-feature interactions to prevent inference attacks; (4) The lack of direct data sharing increases training instability. (5) How to obtain the optimal loss function involves complex optimization problems. Addressing these issues will advance the applicability of differentially private generative models in real-world, vertically partitioned environments.

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